06. INHIBITION AND COLOCALISATIONS

DURATION : 05:42

STUDY SHEET

COURSE SUMMARY

This video explores the complex mechanisms of inhibition and colocalisations in embryonic development, highlighting the importance of left laterality and the crucial role of neural tube closure. Key concepts discussed include the formation of optic vesicles, the interaction between the neural crest and various organ systems such as the hepato-pancreatic system and the heart, and how these interactions influence the evolution of the eye. By integrating principles of electromagnetic theory, the video demonstrates the impact of these processes on embryonic morphology and genomic expression.

INHIBITION AND COLOCALISATIONS IN EMBRYONIC DEVELOPMENT

Lateralised information, sonetic and hutch, is essential in embryonic development, particularly in left-right dynamics. A **left laterality** begins to establish itself, but an inhibitory phenomenon prevents the expression of sonetic and hutch forwards, favouring lateral expression.

This inhibition is linked to the **non-closure of the neural tube**. Without this inhibition, a cyclops eye could develop, as it is the central inhibition that allows the formation of two small lateral vesicles. When the neural tube is not closed, the eye appears around day 22, and the closure of the anterior neuropore occurs around day 24 or 25.

At this stage, so-called **"negative" information** emerges from the neural crest, which shows colocalisations with information from the **hepato-pancreatic** system and the **heart**. The heart receives important information from the neural crest, as does the throat. These colocalisations are expressed at the level of the eye, involving coding genes in different spaces, thus illustrating systems of colocalisation of genetic information.

An **autocorpse** represents a genomic function, acting as a reference for the body. This electrical field, resulting from a polarity, is fundamental. According to **Maxwell's law**, every electrical field generates an electromagnetic field. Thus, each individual receives specific information.

In the anotocorte, the cells at the front receive information called sonetic HH. If this growth is slowed down, an acceleration occurs, leading to an expansion of the epithelium, linked to genetic processes. However, as long as the neural tube is not closed, a small space persists.

Without inhibition, the eye develops centrally, leading to the formation of a cyclops eye. The closure of the neural tube is crucial for the neural crest to express laterally. Once closure is achieved, negative inhibition influences the eye, in connection with **thyroid**, **cardiac**, and **hepatopancreatic** colocalisation phenomena. These large organs have a direct impact on the eye, which is strongly connected to the **choroid**, a vascular mesenchymal tissue.

It is important to consider how the eye interacts in space. The great flexion of the embryo is linked to the appearance of the **optic vesicles**, which form due to growth and this flexion. This flexion is induced by the **primitive aortic vascular axis**, around which the vascular system coils.